

Performance Comparison Between Isolated Amplifiers and Isolated Modulators

Table 1

Category	Isolated Amplifier	Isolated Modulator
Sample resolution	11 bits (bandwidth = 100kHz)	> 14 bits achievable, trade-off between resolution and bandwidth or latency
Latency	2 μ s to 3 μ s (fixed)	< 1 μ s achievable, trade-off between resolution and bandwidth or latency
Bandwidth	up to 300kHz	> 1MHz achievable, trade-off between resolution and bandwidth or latency
Accuracy and drift performance	High	Very high
Number of components needed	More	Less

bitstream of 1's and 0's at a much higher frequency (up to 20MHz). The average time of this bitstream output is proportional to the analogue input voltage. The measured signal is reconstructed with a digital filter inside the microcontroller families such as the TMS320F2807x and TMS320F2837x, DSP, or field-programmable gate array (FPGA).

Performance comparison between isolated amplifiers and isolated modulators

In an isolated amplifier-based solution, the measured analogue signal undergoes several analogue-to-digital and digital-to-analogue conversions. The stages within the isolated amplifier, differential-to-single-ended stage,

Performance Trade-Off Between ENOB and Settling, Latency or Bandwidth for the AMC1306 at CLKIN=20MHz using a Sinc³ Filter

Table 2

OSR	ENOB (Bits)	Settling (μ s)	Latency (μ s)	Bandwidth (kHz)
8	4.65	1.2	0.6	1250
16	7.57	2.4	1.2	625
32	10.02	4.8	2.4	312.5
64	12.3	9.6	4.8	156.25
128	13.51	19.2	9.6	78.13
256	14.11	38.4	19.2	39.06
512	14.39	76.8	38.4	19.53

and ADC reduce overall accuracy and noise performance and increase latency.

The fixed low-pass filter implementation in the output stage of the isolated amplifier limits the signal

bandwidth. An external op-amp-based circuit used for differential-to-single-ended conversions can be used to create an active low-pass filter to further limit the signal bandwidth and thereby improve noise performance.

The isolated amplifier has a fixed latency. Isolated amplifier-based solutions are widely used because of familiarity and relative ease in implementation. As shown in Fig. 2, the measured analogue signal in an isolated modulator-based solution undergoes only one analogue-to-digital conversion. This solution eliminates the need for a differential-to-single-ended stage, thereby reducing the number of components and solution size.

The ADC used in an isolated amplifier based solution, which in many situations limits the maximum achievable sample resolution and accuracy, is not needed anymore. This isolated modulator-based approach has improved signal-noise performance, overall accuracy and can achieve higher signal bandwidth and lower latency than an isolated amplifier-based solution.

Isolated modulators provide a much faster digital bitstream output, typically up to 20MHz. The sigma-delta filter module (SDFM) module inside the microcontroller families (such as the TMS320F2807x and TMS320F2837x) provides an easy way

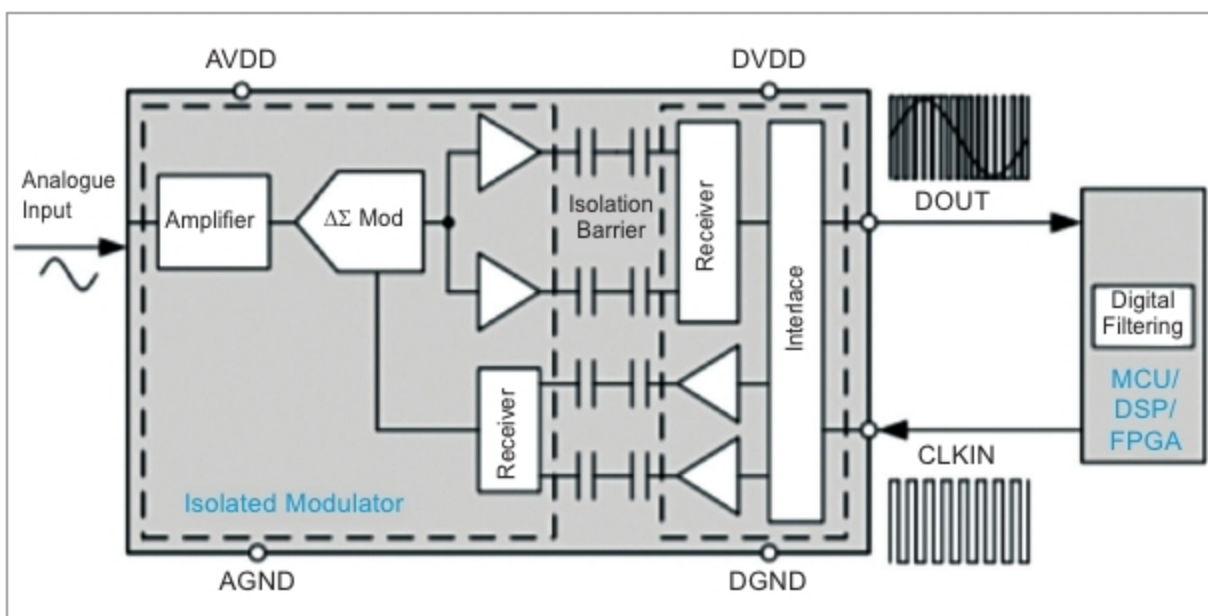


Fig. 2: Isolated modulator implementation

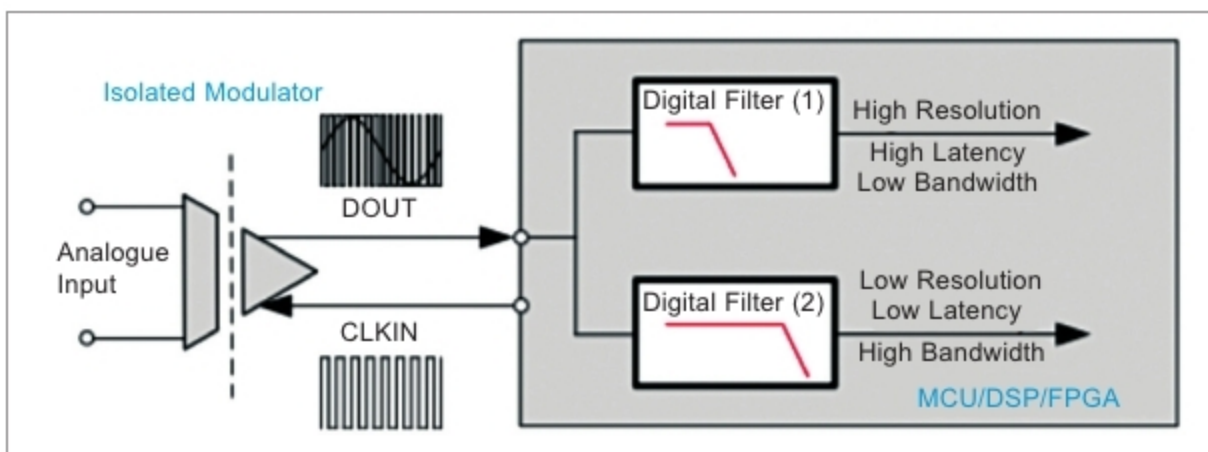


Fig. 3: Implementation of two digital filter in parallel